

Lecture 2: Analytical Methods

Introduction to Data Analysis Techniques

1 Lagrange Interpolation

Definition 1.1 (Interpolation). [def:interpolation] Methods to find a smooth function $f(x)$ that passes through a given set of points $(x_0, y_0), (x_1, y_1), \dots, (x_n, y_n)$

Theorem 1.1 (Unique Existence of Lagrange Interpolation). [thm:lagrange-existence-uniqueness] For any set of $n + 1$ distinct points $(x_0, y_0), (x_1, y_1), \dots, (x_n, y_n)$, there exists a unique polynomial of degree at most n that passes through all these points.

Convention 1.1 (Lagrange Interpolation Method). [conv:lagrange-interpolation-method] Given a set of $n + 1$ distinct points $(x_0, y_0), (x_1, y_1), \dots, (x_n, y_n)$. We may construct the following system of equations:

$$\left\{ \begin{array}{lcl} a_0 + a_1x_0 + a_2x_0^2 + \dots + a_nx_0^n & = & y_0 \\ a_0 + a_1x_1 + a_2x_1^2 + \dots + a_nx_1^n & = & y_1 \\ \vdots & & \vdots \\ a_0 + a_1x_n + a_2x_n^2 + \dots + a_nx_n^n & = & y_n \end{array} \right. \implies \begin{bmatrix} 1 & x_0 & x_0^2 & \dots & x_0^n \\ 1 & x_1 & x_1^2 & \dots & x_1^n \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & x_n & x_n^2 & \dots & x_n^n \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \\ \vdots \\ a_n \end{bmatrix} = \begin{bmatrix} y_0 \\ y_1 \\ \vdots \\ y_n \end{bmatrix}$$

According to the **Theorem in Linear Algebra**, which states that a system of n linear equations with n unknowns has a unique solution if $\det A \neq 0$, where A is the coefficient matrix.

Since we ask all points in the set to be distinct, we can know that $\det A \neq 0$. Therefore, there exists a unique solution $\vec{a}$

So, we can find a unique polynomial $f(x) = a_0 + a_1x + a_2x^2 + \dots + a_nx^n$ that passes through all the given points.